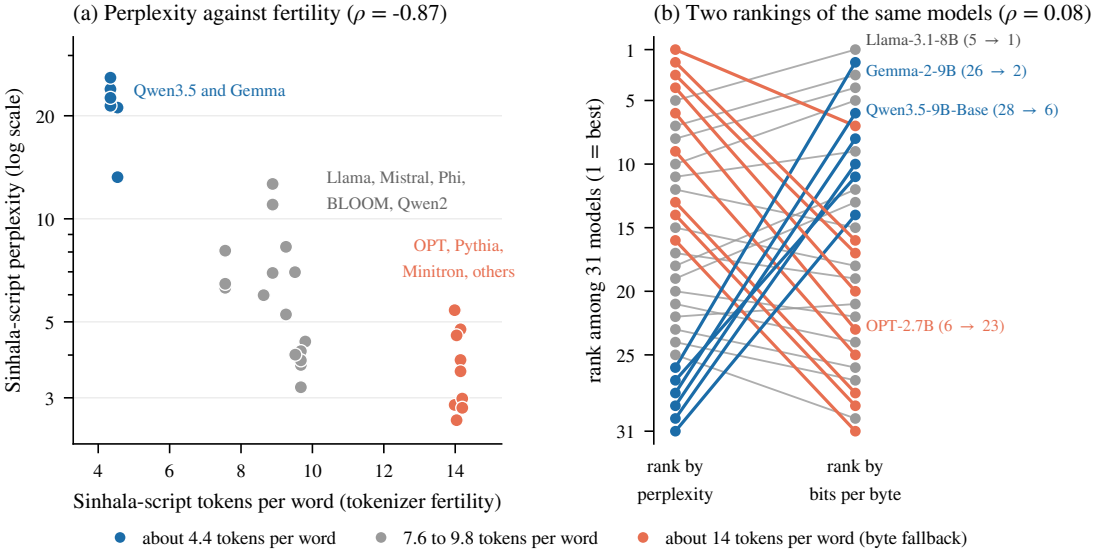
(a) Perplexity against fertility (ρ = -0.87)
Sinhala-script perplexity (log scale)
20
10
5
3
Qwen3.5 and Gemma
Llama, Mistral, Phi,
BLOOM, Qwen2
OPT, Pythia,
Minitron, others
4
6
8
10
12
14
Sinhala-script tokens per word (tokenizer fertility)
(b) Two rankings of the same models (ρ = 0.08)
rank among 31 models (1 = best)
1
5
10
15
20
25
31
Llama-3.1-8B (5 → 1)
Gemma-2-9B (26 → 2)
Qwen3.5-9B-Base (28 → 6)
OPT-2.7B (6 → 23)
rank by
perplexity
rank by
bits per byte
about 4.4 tokens per word
7.6 to 9.8 tokens per word
about 14 tokens per word (byte fallback)